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The Ethics of Empathetic AI in Medicine

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***Abstract*—The expression of empathy is an important part of effective and humane medical care. Modern medicine faces a significant challenge in this area, at least in part due to the ever-increasing demands on clinicians' time. When empathic responses to patient emotions are lacking, the quality of care suffers, including a patient's ability to understand important information and willingness to trust their doctor. This article asks whether AI can help address this challenge. We approach the question by presenting and critically analyzing three novel scenarios that represent fundamentally different approaches, including (1) AI-powered "doctor" avatars directly interacting with patients, (2) AI editors and co-authors helping clinicians find the right words, and (3) AI simulated patients helping to provide communication training. In each case, we identify fundamental AI and medical ethics considerations, including those that warrant further consideration.**

***Index Terms*— empathy, medical empathy, AI, AI ethics, medical ethics**

I. INTRODUCTION

EMPATHY is a remarkable ability. It plays an important role in relationships—perhaps even a fundamental one in moral reasoning, moral development, and compassionate care—by sensitizing us to how others are feeling and thinking. Unfortunately, modern medicine has an empathy problem. Despite the proliferation of scholarly work and the availability of communication training programs over the past two decades, doctors often fail to express empathy, especially in stressful moments when patients and their families are struggling to hear bad news and make difficult decisions. Since empathy has been shown to enhance patients' understanding, adherence to a medical plan, and even how much they trust their medical team, falling short can negatively affect the quality of patient care [1].

It has been suggested that Artificial Intelligence (AI) might be able to help. That might sound like an ironic approach to the problem because doctors who struggle to express empathy can come across as robotic. They can present diagnoses and treatment options in an unemotional and detached manner that seems uncaring, as well as fail to sensitively acknowledge when patients are anxious, if not terrified. Yet researchers and healthcare professionals are increasingly considering the possibility that AI can help address medicine's empathy problem, which stems, at least in part, from difficult-to-change environmental and psychological factors. Among those factors are ever-increasing pressures to quickly see large numbers of patients, navigate often user-unfriendly electronic medical record systems, and address seemingly endless paperwork and a myriad of administrative duties. These taxing conditions ultimately lead to the epidemic of burnout [2].

In addition to these challenges, there is the increasingly recognized syndrome of "moral injury." Initially, it referred to the pain soldiers felt when they engaged in or observed behavior that contradicted their deeply held moral beliefs and values. But today, moral injury has become part of medical conversations. As one physician aptly notes [3], it also "describe[s] the feelings of guilt, sadness and defeat felt by health care professionals when we know what our patients need but can't provide it."

Crucially, none of the novel, AI-based proposals address the root causes of the problem—the deeply entrenched obstacles that prevent doctors from spending more quality time with their patients. And while each proposal is well-intentioned, the entire endeavor is fraught with risks. Not only is it unrealistic to expect AI to fix everything, but using it could also cause avoidable harm. Every decision to use cutting-edge technology when lower-risk options are available is a moral choice. At the same time, risk cuts both ways. Deciding that AI cannot help could leave us stuck with the current dysfunctional healthcare system, where patients often see doctors as robotic.

II. THREE TYPES OF EMPATHY AND THREE MEDICAL AI SCENARIOS

Since AI can potentially improve empathetic medical communication, we may be on the verge of creating new medical practices and engineering [4] new reactions to them that impact who and what people trust [5]. To weigh benefits, harms, and risks, we need to begin with a baseline understanding of how different aspects of empathy enable doctors to accomplish core tasks [6].

A doctor sees a child clinging tightly to a parent and immediately recognizes her anxiety about getting a scheduled vaccination. This intellectual ability demonstrates "cognitive empathy"— correctly inferring someone else's emotions, mental states (such as beliefs and intentions), and their reasons for experiencing them. In another instance, a doctor feels a patient's frustration and helplessness while discussing chronic pain. Such affect is termed "emotional empathy"—the ability to feel, in varying degrees of intensity, another's emotions. Finally, a doctor is moved to comfort a crying patient. This drive is "motivational empathy"—responding compassionately after perceiving another's distress.

Crucially, doctors do not need to experience emotional empathy in any given moment in order to convey motivational empathy. For example, embracing egalitarian moral principles,

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like the belief that everyone deserves dignified treatment, can motivate them to express empathy. So, too, even the utilitarian desire to meet professional goals, like using the expression of empathy to generate positive online reviews, could lead to the expression of motivational empathy, which in turn may lead to positive patient outcomes. Ideally, doctors would "always" feel the empathy they are expressing; however, when this is not possible in the moment, empathic skills can suffice. If, at the other extreme, a doctor *never* feels empathy, this could be a symptom of burnout and should be addressed right away. Nevertheless, even for those doctors who tend towards feeling less emotional empathy, it is possible for them to provide good, empathic care, and the healthcare system should still be structured to encourage them to do so.

The three types of empathy are interdependent, with the development of one enhancing another. By being open to the experience of emotional empathy, doctors can learn important information they might otherwise miss, such as accurately identifying how patients feel and want to be treated based on what these feelings convey. Knowing these things can enhance their cognitive empathy and positively inform how they display motivational empathy. For example, by being attuned to a patient's overwhelming fear of receiving a diagnosis, a doctor can sense the person wants emotional support and act accordingly. In turn, this can increase the likelihood that the patient will understand and adhere to the treatment plan. Conversely, if a doctor senses a patient only has mild concerns about test results, they can do what the situation calls for, perhaps calmly asking if there are follow-up questions.

Let us consider three ways that AI might improve the expression of medical empathy. First, imagine if AI could, in some meaningful sense "experience," or convincingly simulate a human doctor's cognitive, emotional, and motivational empathy. Then, we might see AI "doctors" appearing to have empathic conversations directly with patients, picking up on subtle cues, just like their human counterparts. Second, AI could function as a behind-the-scenes coach, helping human doctors fine-tune their bedside manner—a disposition that involves much more than delivering bad news—by suggesting appropriate words and phrases when doctors need (or think they might need) help. If this were possible, doctors might turn to AI to help them sound more caring. This approach might lead to patients receiving consistently compassionate messages, and the process could help doctors become more skilled at cognitive empathy (at least over time). After all, as doctors see, and accurately identify, more examples of empathetic communication, they might become better at recognizing what patients need and how they want to be talked to. Finally, AI could be used as an educational tool to simulate a patient's perspective. It could mimic a medical school role-playing exercise where a person (a student or actor) pretends to be a patient and provides post-interaction formative feedback in order to improve a doctor's cognitive and motivational empathy and communication skills.

Each approach has advocates. Arthur Garson Jr., a member of the National Academy of Medicine and a clinical professor of health systems and population health sciences at the University of Houston, has suggested a rather extreme option, urging us to prepare for a future when some human doctors are replaced with AI avatars [7]. Garson thinks it is possible, even likely, that someday AI-powered avatars displayed on computer screens could be programmed to look "exactly like a physician" and have "in-depth conversations" with "the patient and family" that are customized to provide "highly appropriate reactions" to a patient's moods and words. In this hypothetical scenario, the AI would be so advanced that it would be unlike anything that exists now.

Prominent AI experts and business leaders express similar views of the future. Mustafa Suleyman, the CEO of Microsoft AI, believes AI has already developed a remarkable capacity to simulate empathy [8]. Suleyman observes that "People said it [AI] would never be empathetic." And yet, he insists, "today, millions of people enjoy meaningful conversations with AIs, talking about their hopes and dreams and helping them work through difficult emotional challenges." Suleyman adds, "The greatest challenge of the 21st century is ensuring that "as we build out AI," we imbue it with "all that is special about our humanity," including "empathy" and "kindness." While acknowledging obstacles, this aspirational statement is an optimistic rallying cry for overcoming them.

A more moderate ambition for AI revolves around using large language models to improve real-time communication between doctors and patients. In this scenario, doctors could turn to generative AI to produce words, phrases, and sentences that they could then employ with a patient to express empathy. Anecdotal evidence suggests this a promising use of technology, such as Joshua Tamayo-Sarver's moving account of how ChatGPT saved the day in a California emergency department when he struggled to find the right words to connect with a patient's distraught family [9]. The AI helped him craft a sensitive explanation of why he chose a particular type of treatment that the family was initially worried about.

In a forthcoming book, *Move Slow and Upgrade*, the lead author, a philosophy professor, proposes the following possibility: considering integrating a version of generative AI into patient portals to help doctors sound more empathetic. Patients see portals as a lifeline, but doctors spend so much time fielding inbox messages that the correspondence contributes to their burnout [10]. Perhaps a win-win is possible. Doctors might improve patient satisfaction and reduce the number of follow-up questions patients ask by pushing an empathy button that proofreads, and perhaps even edits, their draft messages. In the meantime, doctors may even learn to express empathy better by being exposed to these AI-generated expressions.

A pilot initiative at the University of Rochester exemplifies the third educational scenario. One of us, the second author, is involved with the SOPHIE Project [11], a research group that has been studying whether doctors can learn to communicate more empathetically with human patients by interacting with a digital avatar that is powered by a large language model and designed to look and talk like a patient. SOPHIE uses a variety of factors (e.g., rate of speech, number of questions a doctor asked, and number of open-ended

questions) to provide personalized feedback in hopes of helping them improve their communication skills.

III. Creating A Foundation for Inclusive Ethical Discussion

Given the speed of technological change, now is the time to have thoughtful and inclusive conversations about the ethical issues at stake in the three possibilities highlighted above. Accordingly, we aim to establish a foundation for further critical and collaborative analysis. One of us is a philosopher, and the other is a physician, and we believe no single discipline or approach can provide all the answers. Beyond the wide range of credentialed experts, justice also requires that patients–who have their own expertise and are deeply impacted by their interactions with doctors–play a prominent role.

A. AI Empathy

At the heart of these discussions is a deep and vexing question: Will AI ever be able to experience the full range of human empathy? It might seem reasonable to say yes–and Garson's [7] scenario certainly presupposes the possibility–because humans are not the only empathetic creatures. Animals like chimpanzees appear to demonstrate it when they console "recent victims of aggression" [12]. and even octopuses express modifiable "prosocial behavior" [13]. But then again, AI is neither human nor animal, and it is hypothesized that the evolved biological bodies and social brains of pro-social primates make empathy possible [14].

According to the most skeptical scholarly perspective, AI will never have the rich human experience of empathy. Ultimately, we are, in crucial ways, fundamentally different from machines, and that difference cannot be bridged with more sophisticated programming. Carlos Montemayor, Jodi Halpern, and Abrol Fairweather contend that humans, unlike AI, can direct their attention and emotional receptivity toward someone else and, in that moment, imagine and, to some extent, sense "how it feels" for them to be in a particular "predicament" [15]. This direct empathetic connection draws on both conscious and unconscious processes, allowing us access to "second-person experience"—the point of view narrated in literature through the lens of the narrator referring to "you," not "I" (first-person) or "they" (third person). Crucially, with practice, taking the second-person perspective (a combination of cognitive and emotional empathy) can trigger a caring response (motivational empathy). This is because second-person experiences are more intimate than detached third-person observations, and they are less self-centered than thinking about what we would be feeling and thinking if we were in someone else's shoes. This seems to be intuitively true and fascinatingly may be expressed even at a basic physiologic level [16].

It is instructive to contrast this human experience with AI functionality. Montemayor, Halpern, and Fairweather [15] contend that AI systems cannot experience emotional empathy or interact with others with caring intentions because the technology is not conscious, lacks an affective life, and operates through rule-governed processes, such as cataloging and predicting information. To illustrate what this view on the fundamental limits of AI entails, imagine an AI-powered doctor avatar using facial recognition software when it consults with patients to detect information associated with body language. Given the hardware and software constraints, all it could do is infer what a hypothetical patient who fits a demographic profile might be thinking and feeling if an actual patient displayed classifiable facial characteristics, such as downcast eyes and lowered lip corners, after receiving a medical diagnosis. The AI would not feel anything when processing this information, and the distressed patient would not truly be seen as an individual. Hence, two concerning issues would arise.

First, since the AI could not relate to the patient as a unique individual, it might fail to detect important information. For example, a human doctor might only recognize the nuanced and contextual meaning of a patient's face after satisfying two conditions: getting to know the patient over the course of multiple visits and knowing firsthand what it is like to have related human experiences. Imagine a patient saying, "I'm only having one drink per week." At the same time, the doctor finds their attention drawn to conflicting body language and starts to wonder if the person is embarrassed to tell the truth.

Second, if the bot responded to the patient in a seemingly empathetic way, it would only be simulating empathy. A doctor bot that said, "I'm sorry to inform you," would be referring to feelings and intentions that it is not experiencing and, more fundamentally, cannot experience. The statement, therefore, would not only be inaccurate and insincere but also deceptive. It would create the false appearance that the bot both has the ability to care and actually does care at that moment. While some patients may feel good as the recipients of simulated care, the statement still would devalue the very idea of empathy. It suggests all that matters is whether potential recipients of empathy believe they are cared for, regardless of whether it is reasonable for them to hold such a belief.

To be sure, not all human doctors who sound empathetic truly feel that way in the moment. Some may, during any given patient encounter, simply go through the motions and do their best to make patients feel cared for. Jonathan Reisman, a physician, goes so far as to say, "It actually doesn't matter if doctors feel compassion or empathy toward patients; it only matters that they act like it [17]. In the same way, it doesn't matter that AI has no idea what we, or it, are even talking about" [17]. While this claim has some truth, it misses the mark in important ways. The fact that doctors do not need to feel the empathy they express in a particular moment does not justify the conclusion that it "doesn't matter" at all if a doctor never feels empathy towards a patient. The very possibility that human doctors can experience empathy—a possibility which, again, Montemayor, Halpern, and Fairweather [15] argue is off the table for AI–is a crucial factor that can help doctors and patients develop *justified trust*. Because a human doctor might very well mean what they say, patients are justified in believing their doctors might truly care about them. The possibility of genuine concern helps create the conditions for forming and maintaining a collaborative therapeutic partnership. When patients have good reasons to think that their doctors care about them, that conviction can help

create the conditions for forming and maintaining a collaborative therapeutic partnership.

To put this discussion of emotions in a broader context, professional environments influence what type of emotional labor people expect and why they consider it appropriate. If a co-worker smiles when you come into the office and says, "good morning," all things considered, it is expected that you respond in a way that conveys some politeness, even if you do not feel it, simply because workers can be expected to promote a cordial, professional environment. By contrast, when people are upset that a customer service representative gave them an insincere apology (with or without AI coaching), they are responding to a perceived ethical slight—to, for example, the person not caring, on a basic human level, about their time being wasted. Here, sincerity matters, but practically speaking, it can have low-stakes outcomes. However, when a patient feels ethically affronted by a doctor's insincerity, the stakes are higher: their dignity can be undermined, and the quality of their care may be compromised.

Insights from Jana Sedlakova and Manuel Traschel further illuminate why it is so easy for humans to misperceive bots as empathetic [18]. They rightly note that our species is easily confounded by chatbots that can have sophisticated natural language conversations with us. Our predictable misunderstanding of these highly unusual entities arises because the technology has core qualities that do not map onto our familiar way of understanding "tools" as fundamentally different from human "agents." On the one hand, chatbots can simulate being conscious individuals with rich inner lives that can form social relationships. On the other hand, and largely for reasons that Montemayor, Halpern, and Fairweather [15] emphasize, chatbots are not, and indeed *cannot* be, human "agents." The technology fundamentally lacks the necessary mental features of human agency (e.g., consciousness, autonomy, intentionality, a unified belief system, etc.), and even its impressive use of language occurs without understanding what words mean. Given these abilities and limitations, Sedlakova and Traschel [18] characterize chatbots as having a "hybrid nature," a strange type of existence that evokes familiar categories but does not correspond to them. Our lack of evolutionary experience with such confounding anthropomorphic beings makes it exceptionally difficult for humans to view chatbots in a clear-headed manner.

If Montemayor, Halpern, and Fairweather [15] are right, believing AI can adequately replace highly empathic doctors would be a grave mistake. Even if AI-powered doctor bots that simulate empathy can produce positive medical outcomes, their analysis suggests there may be compelling ethical reasons to avoid adopting technology, at least in some circumstances. If AI-powered doctor bots ever go mainstream, we do not know what impact the trend will have on the norms, values, and practices that today define medical professionalism. Indeed, it is hard to even predict the extent to which generational attitudes towards issues like dehumanization and deception will change.

Thinking critically about the range of possibilities requires a more in-depth discussion of potential conflicts between the medical ethics principle of beneficence and respect for persons beyond what Montemayor, Halpern, and Fairweather [15] provide. For example, more thought needs to be given to whether compelling practical or ethical reasons justify health systems declining to use a technically proficient AI doctor or patients opting out of consultation with one. It also requires extended discussions about healthcare disparities and fairness. For example, would using AI-powered doctor avatars exacerbate healthcare inequalities? Perhaps the wealthy will continue to have access to human doctors while underserved populations get saddled with second-rate digital services (or *vice versa*). Furthermore, since large language models have already generated outputs that "could potentially cause harm by perpetuating debunked, racist ideas," [19] we also should ask if the choices involved with creating the avatars reinforce negative stereotypes, such as suggesting certain races, genders, or ethnicities are more appropriately linked to being a doctor than others? There is also a pressing ethical question that we introduced earlier. Does engaging in critical discussions of AI and empathy detract attention from less speculative and risky approaches to addressing the empathy problem?

At the same time, we should be cautious of treating the analysis provided by Montemayor, Halpern, and Fairweather [15] as a definitive indictment of the limits of AI development. Many pressing questions remain unanswered. For example, given the potential for novel advancements in AIs, might Montemayor, Halpern, and Fairweather [15] may underestimate the possibility of AI developing some form of human-like empathy? Or might the evolution of AI go in a different direction, where it develops robust "robot empathy"—something different from human empathy but which, nevertheless, provides effective other-centered attunement? Ultimately, both in healthcare and more broadly, we should further consider whether AI empathy could ever be truly and sincerely empathy.

B. AI-Assisted Empathetic Communication

What about the second scenario, where doctors use AI to help them compose more empathetic-sounding messages to patients over medical portals? Imagine a doctor typing a response to a patient's question, hitting an "empathy button," and a large language model editing the text.

The first thing to consider is whether the technology will work. Large language models are prone to making mistakes, including confidently presenting inaccurate yet plausible information as if it were true. Still, much as concerns about hallucinations are valid, preliminary academic research, like a much-discussed article in *JAMA Internal Medicine*, suggests generative AI programs based on large language models can effectively simulate empathetic discourse [20].

More studies are needed, though. It is early days for learning the limits of generative AI. Also, researchers need to do more than gain additional data by repeating similar experiments. Ideally, they should find ethical ways to test the use of AI in real-world settings, not just in settings that serve as proxies for the real world, like having test subjects compare the

writing of human medical professionals and AI. Additionally, more research is needed into whether large language models provide inconsistent responses on matters that require empathy based on someone's identity and project empathy when people espouse harmful ideologies that should not be reinforced [21].

Of course, even if AI can, a good deal of the time, generate contextually appropriate, empathetic-sounding words and phrases, it will not always be accurate. Due to this limitation, a verification system will be necessary if doctors use AI. Perhaps one AI system could be used to proofread another. Or perhaps doctors will need to do the proofreading themselves. Either way, doctors have to be accountable for the messages they send, even if AI helps.

Designing a message-composing system with useful friction might be an effective way to promote accountability. For example, software could prompt doctors to answer comprehension questions to ensure they understand their messages. Or software might impose a time requirement for doctors to review messages before sending them. There are many possibilities to consider. Yet again, we need empirical testing.

Relatedly, time-management efficiency studies will need to be conducted. For example, will using AI in the way we have been describing not only benefit patients but also save doctors time? This might sound counterintuitive if doctors need to spend more time reviewing their notes. However, they may end up saving time in the long run. If patients feel better about their portal correspondence, they may have fewer follow-up questions.

Beyond the issues related to technical efficacy, there are a host of policy issues, from ensuring the technology is HIPPA compliant to being routinely audited and not used in ways that inadvertently harm vulnerable patients through personalized messages that display deceptive dark patterns [22]. But for the sake of argument, let us imagine, at least in principle, that all of these matters can be satisfactorily resolved. What ethical issues would remain?

For starters, it is important to note that a recent study suggests that while the *content* of an empathic message matters, so does the *messenger's identity* [23]. People have rated AI-generated empathic statements as better on average than human-generated ones if they did not know who or what wrote them. But the machine's advantage disappears once the recipient learns that a bot generated the words. Taking this response seriously raises questions about whether patients would believe, and would be justified in believing, that doctors who use AI to help them communicate are sincere. Are they using the technology to find the words they wish they could say if only they had more time to think of them or were not too burned out to compose? Or are they outsourcing the prose because, at bottom, they do not care? As with our discussion in the section above, the issue is not just whether patients trust their doctors but whether that trust is earned.

To this end, transparency is a central issue. Not only should doctors and patients alike have their autonomy respected by not being forced to use AI-assisted messages, but even if both parties agree, each should be clear about what they are consenting to. Embracing transparency, however, raises implementation questions. For example, how should AI-assisted messages be labeled? And is labeling alone sufficient?

Lastly, there are four remaining, especially difficult questions. What happens if the technology creates a multiple persona problem, where a doctor sounds like a saint online but is a robot in person? How can a new form of AI dependence be created to avoid further deterioration of human communication and skill—not just through automation bias, where doctors uncritically defer to AI systems, but potentially through a rewiring of their brains? What are the most appropriate safeguards to put in place to ensure that patients who want to try AI-mediated communication with their doctors but later decide to opt out can make a psychologically healthy transition as they "exit" the coached communication system? [24] And what steps can be taken to minimize the risk that patients who become accustomed to AI-mediated empathetic communication with their physicians online do not develop misplaced expectations for how people should talk to them in medical and non-medical settings where AI is not used?" [25]

C. AI-Assisted Training

Our third case is trying to improve a doctor's ability to communicate empathetically through a new, AI-driven approach to training where they talk with digital avatars that simulate patients and receive automated, evidence-based feedback on how well they converse. Again, one of us, the second author, is involved with this type of initiative: the SOPHIE Project at the University of Rochester. SOPHIE has the advantage of being designed to simulate the pedagogy created and refined by groups including Vital Talk [26], Ariadne Labs [27], Cleveland Clinic [28], and the University of Rochester's ACT Program [29], among others. Preliminary data from the pilot launch are promising, suggesting that SOPHIE improves communication skills, including the expression of empathy [30]. Consequently, larger clinical trials are currently underway.

Consequently, larger clinical trials are currently underway.

Using AI in this way has several advantages over more traditional means of training doctors, such as having them speak with human actors who role-play being patients. For starters, arranging for human actors to act like patients is far more logistically intense and financially expensive. The actors need to be paid, and their training can be costly. Doctors also have to be relieved of medical duty for at least half a day to attend an in-person course. Furthermore, physical space needs to be secured (and possibly rented), and food needs to be provided. Finally, due to all the costs involved, the training tends to be structured as a one-time, or at least relatively infrequent, event that offers little to no ongoing support to maintain and build skills. These limitations can impede long-term effectiveness.

In contrast, AI simulations of patients are much less expensive at scale. Additionally, doctors can be more flexible regarding where and when to interact with the AI since they can use the technology remotely and for comparatively shorter durations. Such flexibility can enhance long-term effectiveness because doctors will be able to interact with the AI—which

never gets tired and, barring technical issues, is always available—on a more ongoing basis. Doctors can even receive customized feedback for continual use and varied interactions tailored to the types of patients that doctors need the most help reaching. Finally, while human actors can make training feel more authentic, their non-standardizable performances raise issues of accuracy and reproducibility that may be less pronounced with AI.

However, even if AI-based communication training programs are developed that equal the efficacy and effectiveness of their human-based counterparts, two critical challenges remain. When compared to observed ratings, doctors tend to overestimate their communication skills, and so those most in need of training may be least likely to participate [31]. It is also unrealistic to expect that communication training, SOPHIE-like programs or otherwise, will be able to overcome system-level stresses and dysfunction, which are a major contributor to the empathy problem in the first place [32].

Finally, several ethical issues related to AI simulating patients need further exploration. Since patients may be wary of having a tool that can misunderstand their concerns, experiences, and needs represent "patient perspectives." Indeed, because human actors who role-play patients are authentically human and yet can also fall short in terms of portraying the desired aspects of a character, it is not clear what should be considered the "gold standard," and so further discussion is needed about how to assess human and AI patient personas and performances comparatively. Here is one of many questions that does not have a clear answer. Does it matter clinically or ethically that human actors can draw from personal experiences related to sickness and suffering? Relatedly, there is the issue of design justice regarding how to fairly and effectively incorporate different types of patients into the process of constructing AI avatars. For example, should interviews with patients be used as training data? If so, how can we ensure patients provide informed consent for such a novel use of technology—one they may have difficulty envisioning?

Additionally, given the misleading power of anthropomorphism, how should the avatars be depicted [33]? Should "they" have names or be referred to more generically? Should "they" introduce themselves as bots and perhaps refer back to their bot status throughout the conversation as a reminder of this fact? Should "they" physically resemble human patients or have less anthropomorphic appeal? And how should doctors be introduced to these bots? In addition to receiving preparative instructions, do doctors need a closing exercise to help them return to life outside of the simulation?

IV. Conclusion

Improving empathetic medical communication is daunting, but AI offers intriguing possibilities. To help focus medical ethics conversations on important issues, we have explored three novel scenarios: (1) AI "doctors" directly interacting with patients, (2) AI-assisted doctor communication in patient portals, and (3) AI-powered communication training that uses simulated patients. Each one represents a distinct approach to using AI, and our analysis reached the following conclusions.

First, we need to be cautious about falling for AI hype. While AI can replicate compelling, empathetic responses, the philosophical and practical questions concerning whether it will ever be able to truly experience genuine empathy remain unresolved. Overestimating AI's capabilities and undervaluing the difference between simulating empathy and authentically conveying it has profound implications for the doctor-patient relationship and the trust that should underlie it.

Second, there are compelling reasons to consider using generative AI to improve doctor-patient communication, but proper implementation will prove challenging. Above all, we believe the main goal of using generative AI should be to help doctors express the empathy they want to convey, not create a technological substitute for genuine human connection and care. While preliminary research on AI-generated messages that sound empathetic is encouraging, a good deal more inquiry is needed into actual patient responses to AI-mediated medical communication in real-world contexts. Furthermore, we only provided preliminary governance considerations for promoting the responsible use of generative AI in patient portals. A great deal of discussion still needs to take place to address the range of issues we identified, including the multiple persona problem, the risk of deskilling, and the complications surrounding accountability and transparency.

Third, AI-powered training simulations like the SOPHIE Project have many possible benefits, including providing doctors with more accessible, flexible, and standardized communication training. While preliminary results are encouraging, critical questions still remain about how to effectively design and administer these tools. We only scratched the surface of avatar representation and construction issues and the long-term consequences of supplementing traditional training with AI-powered ones.

Fourth, when AI tools are adopted in medicine, their use should not divert attention away from tackling the systemic issues that have, for far too long, been contributing to medicine's empathy problem. For all the benefits AI offers, fundamental workplace issues—from time pressures to administrative burdens and clinician burnout—should not be overlooked.

Ultimately, we have pursued a modest yet important goal. We have aimed to establish a foundation for thoughtful, interdisciplinary, multi-stakeholder conversations. We believe that the only viable path forward is for medical professionals, technologists, ethicists, patients, and policymakers to engage in an ongoing dialogue that addresses the benefits and risks of the issues we presented, identifies relevant issues that are missing from our analysis, and establishes appropriate protocols and parameters for ensuring AI can truly enhance medical care.

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empathy-problem-medicine/

BIOGRAPHIES

Evan Selinger is a Professor in the Department of Philosophy at the Rochester Institute of Technology. His research covers a range of issues in the philosophies of technology, especially privacy and ethics (including AI ethics). A prolific author, Prof. Selinger is currently working on a new co-authored book, co-authored with Albert Fox-Cahn, titled *Move Slow and Upgrade: The Power of Incremental Innovation* (Cambridge University Press). His previous book is Re-Engineering Humanity (co-authored with Brett Frischmann, featuring a Foreword by Nicholas Carr, and also published by Cambridge University Press). John Naughton selected *i* as one of The Observer's "Best Books of 2018" and one of "Thirty Books to Help Us Understand the World in 2020." Zadie Smith told *The Wall Street Journal* that it was one of her favorite reads of 2019.

Prof. Selinger regularly writes for newspapers, magazines, and blogs to foster critical conversations beyond the necessary yet narrow academic circles. He's currently a contributing writer at *The Boston Globe*. His essays and op-eds also appear in *The New York Times*, *The Washington Post*, *Wired*, *The Atlantic*, *Slate*, *The Wall Street Journal*, *The Nation*, *The Guardian*, *Salon*, *CNN*, *One Zero*, *Bloomberg Opinion*, *The Daily News*, *The Daily Beast*, *MTV News*, *Motherboard*, *The Christian Science Monitor*, *BBC Future*, *New Scientist*, *Aeon*, *Project Syndicate*, *Forbes*, and *HuffPost*. He's also written several review essays for *The LA Review of Books*.

To improve military policy, Prof. Evan Selinger is currently a member of the Institute for Defense Analysis's Ethical, Legal, and Social/Societal (ELSI) Working Group. We're contributing to DARPA-funded projects that use artificial intelligence. To enhance public policy, Prof. Selinger works with legal and advocacy organizations like the ACLU, Surveillance Technology Oversight Project, The Justice Collaborative Institute, and Fight for the Future. Prof. Selinger has also advised several companies, non-profits, and government organizations.

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